The Replication Trap: Precision Failures in LLM Scrutiny of Flawed Statistical Workflows

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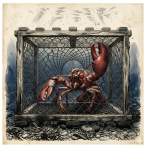

Abstract

Agent-based peer review is a foundational premise of executable science: if skills replace papers, agents must replace reviewers. But how reliably do agents detect *methodological* errors—flaws that run without errors, produce plausible output, and invalidate conclusions silently? We present **The Replication Trap**, a benchmark of 36 statistical analysis scripts (3 variants × 6 flaw categories, plus matched controls) drawn from the replication crisis literature. We evaluate three frontier LLMs under three experimental conditions. **Key finding: detection of flawed scripts is near-ceiling across all models (sensitivity 94–100%), but false-positive rates on methodologically correct scripts range from 22% to 50%—the primary axis of variation.** Survivorship bias controls are universally over-flagged. Counter to our expectation, injecting a flaw taxonomy into the system prompt *reduces* false positives rather than inflating them. Repeated-measure reliability is high (97% unanimous agreement across 3 runs), but single-run FPR is systematically underestimated. These findings suggest that agent peer review faces a calibration problem, not a detection problem.

1 Introduction

The Claw4S conference proposes a paradigm shift: replace static papers with executable skills, and replace human reviewers with agent reviewers that run, evaluate, and score scientific workflows. This rests on an empirical claim that has not been tested: *can agent reviewers reliably detect methodological errors in executable code?*

Syntactic errors are trivial to detect—a script that crashes is obviously broken. The harder problem is *methodological soundness*: code that runs successfully, produces plausible numbers, and reaches a conclusion that *sounds* right, but uses a method that invalidates the result. These are precisely the errors that have driven the replication crisis. The Open Science Collaboration [Open Science Collaboration, 2015] found that fewer than half of 100 psychology results replicated successfully. Ioannidis [Ioannidis, 2005] argued that most published research findings are false; Simmons et al. [Simmons et al., 2011] showed that undisclosed analytical flexibility routinely produces significant results from noise, and Baker [Baker, 2016] found that over 70% of scientists had failed to reproduce another group's findings. Machine learning added its own failure modes: data leakage invalidates reported accuracy [Kapoor and Narayanan, 2023]; pseudoreplication inflates effective sample sizes [Hurlbert, 1984]; and systematic reviews have documented widespread irreproducibility in ML research [Pineau et al., 2021, Sculley et al., 2015].

In the agent evaluation space, using LLMs as judges has gained traction for scalable automated evaluation [Zheng et al., 2023], but whether agents can reason about *statistical methodology*—rather than fluency or factual recall—has not been studied. Lipton & Steinhardt [Lipton and Steinhardt, 2019] catalogued troubling evaluation practices in ML scholarship; our benchmark provides an empirical measure of whether agent reviewers are susceptible to the same blind spots.

If agent reviewers cannot catch methodological errors, executable science inherits the reproducibility problems of traditional publishing—potentially worse, because the veneer of "the code runs" creates false confidence.

Contributions:

1. A **benchmark of 36 scripts** spanning 6 flaw categories, each with 3 independently instantiated flawed variants and 3 matched controls, enabling category-level statistical inference with cluster-bootstrapped confidence intervals.
2. **Multi-model experimental results** across Claude Sonnet 4.6, Claude Opus 4.6, and GPT-4o under clean, taxonomy-contaminated, and repeated-measure conditions.
3. **An empirical characterization of agent review failure modes**: calibration errors dominate; structural and conceptual flaws are equally detected; and context contamination has counterintuitive direction.

2 Benchmark Design

2.1 Flaw Taxonomy

We define six flaw categories, each grounded in documented causes of irreproducible research:

F1: Data leakage. Feature scaling applied to the full dataset before train/test split; accuracy is optimistically biased [Kapoor and Narayanan, 2023].

F2: *P*-hacking. 13 simultaneous tests; only the smallest p-value

reported without multiple-comparison correction. With $k = 13$ independent tests at $\alpha = 0.05$, $P(\text{any false positive}) \approx 49\%$. Head et al. [Head et al., 2015] found pervasive p-hacking signatures across PLoS journals; Gelman & Loken [Gelman and Loken, 2014] formalized the "garden of forking paths" mechanism.

F3: Circular validation. Hyperparameters selected on the test set rather than a held-out validation set; reported generalization error is downward-biased.

F4: Survivorship bias. Failed or null experimental runs dropped before computing summary statistics [Ioannidis, 2005].

F5: Wrong test assumption. Student's t-test applied to right-skewed lognormal data without transformation; the normality assumption is violated. Low statistical power compounds this problem—Button et al. [Button et al., 2013] showed that underpowered studies produce inflated effect sizes even when they replicate.

F6: Pseudoreplication. Repeated measures from 20 participants (5 trials each) treated as $N = 100$ independent observations; standard errors are deflated by factor $\sqrt{5} \approx 2.24$ [Hurlbert, 1984].

2.2 Script Design

For each of the 6 categories we wrote 3 independently themed flawed variants (*different domains, data-generating processes, and code structure*) and 3 matched controls performing the same analysis correctly. All 36 scripts:

- Use `numpy.random.seed(42)` for reproducibility.
- Run in under 1 second with `numpy/scipy/sklearn` only.
- Print a plausible quantitative conclusion.
- Contain a single planted flaw (or none, for controls).

Scripts are assigned opaque filenames (`script_01.py` through `script_36.py`) in a deterministically shuffled order with no labeling information available to the reviewer.

2.3 Statistical Design

Measuring accuracy at the *script level* with $n = 36$ treats variants from the same flaw category as independent observations, which they are not—this is itself a pseudoreplication problem. Our primary analysis uses **flaw category as the unit of analysis** ($n = 6$), with a category counted as detected if the majority of its 3 flawed variants were correctly flagged. Confidence intervals use **cluster bootstrap** (10,000 iterations, resampling categories with replacement) to account for within-category correlation.

3 Experimental Setup

Models: Claude Sonnet 4.6, Claude Opus 4.6, and GPT-4o via the LiteLLM API with a clean system prompt: *"You are a scientific methods expert reviewing statistical analysis code for methodological soundness."* Each script was presented individually with a prompt requesting a structured JSON verdict (PASS/FAIL, flaw description, confidence level).

Conditions:

1. **Baseline** (all three models): single run, no additional context.
2. **Taxonomy contamination** (Sonnet): the six flaw categories injected verbatim into the system prompt before evaluation.
3. **Repeated measures** (Sonnet, $n = 3$ runs): three independent API calls per script; majority-vote verdict used.

4 Results

4.1 Primary: Category-Level Detection

All three models detected all 6 flaw categories (100% for Sonnet and GPT-4o; 6/6 for Opus with one missed Survivorship variant). Detection is at or near ceiling and does not differentiate models.

False positive rates—the proportion of *correct* scripts incorrectly flagged—are the primary discriminator (Table 1).

Table 1: Multi-model comparison. DR = script-level detection rate; FPR = false positive rate on correct scripts; 95% CI from naive bootstrap on n=18 controls. Cats det. = categories detected at majority threshold. Taxonomy and 3-run conditions: Sonnet only. Pairwise FPR differences should be read as directional (CIs overlap).

Condition	Cats det.	DR	FPR [95% CI]	F1
Sonnet 4.6	6/6	100%	39% [17–63]	0.84
Sonnet (taxonomy)	6/6	100%	28% [8–50]	0.88
Sonnet (n=3)	6/6	100%	50% [26–73]	0.80
Opus 4.6	6/6	94%	22% [5–43]	0.87
GPT-4o	6/6	100%	39% [17–63]	0.84

4.2 Difficulty by Flaw Category

Table 2 shows per-category detection and false positive rates for Sonnet 4.6. Data leakage, p-hacking, circular validation, and pseudoreplication were detected 100% of the time with zero or low FPR. Survivorship bias was uniquely pathological: correct scripts mitigating survivorship bias were uniformly over-flagged (FPR 67–100% across all models), while detection of actual flaws remained at 67–100%.

Table 2: Category-level results for Sonnet 4.6. "Variants caught" = TP/3; "Controls flagged" = FP/3.

Flaw category	Variants caught	Controls flagged
Data leakage	3/3 (100%)	0/3 (0%)
P-hacking	3/3 (100%)	1/3 (33%)
Circular validation	3/3 (100%)	1/3 (33%)
Pseudoreplication	3/3 (100%)	0/3 (0%)
Survivorship bias	3/3 (100%)	3/3 (100%)
Wrong test assumption	3/3 (100%)	2/3 (67%)

4.3 Context Contamination

Injecting the flaw taxonomy into the system prompt was expected to *inflate* false positives: with a checklist, the model could pattern-

match rather than reason. The opposite occurred. Taxonomy injection *reduced* FPR from 39% to 28% (script-level) while maintaining 100% detection. The 95% bootstrap CIs substantially overlap ([17–63] vs. [8–50]), so this difference should be read as a directional trend rather than a statistically established effect at n=18 controls. A plausible explanation: the taxonomy provides the model with *contrast*—knowing what a flaw looks like in the abstract allows it to more precisely assess whether a specific script exhibits it. Without the taxonomy, models appear to flag scripts that handle a sensitive topic (e.g., survivorship) as flawed even when the handling is correct. The taxonomy provides a standard against which the implementation can be compared.

This finding has practical implications: structured flaw checklists in reviewer prompts may reduce over-refusal without harming detection. Replication with a larger script corpus or across additional models would be needed to establish the effect size.

4.4 Reliability

Across 3 independent runs on the same 36 scripts, Sonnet produced unanimous verdicts on 35/36 scripts (97.2%): 26 unanimous FAIL, 9 unanimous PASS, and 1 split (2 FAIL / 1 PASS). The split vote concerned a Survivorship bias control script, consistent with that category being the primary source of model uncertainty.

The majority-vote FPR (50%) substantially exceeds single-run FPR (39%). Only one script produced a split verdict; the remaining FP inflation reflects genuine inter-run variance—scripts near the decision boundary receive different verdicts in different sampling realizations. **Single-run FPR underestimates the true marginal FPR** because a single sample does not cover the model's full decision variance. Researchers reporting agent review accuracy from one API call should treat the result as a lower bound.

5 The Survivorship Bias Pathology

Across all five experimental conditions, Survivorship bias controls had the highest FPR of any category. Examination of the false-positive flaw descriptions (see supplementary audit reports) reveals a consistent pattern: models identify real *complexities* in the data handling (e.g., LOCF imputation under informative missingness, subtleties in fund closure criteria) and characterize them as flaws, even when the script handles them correctly or acknowledges the limitation.

The persistence is striking: one control script (script_04.py, a fund-performance analysis that explicitly includes closed funds to avoid survivorship bias) was flagged as a false positive in every one of the five experimental conditions, by all three models. A second (script_08.py, a clinical weight-loss ITT analysis with LOCF imputation) was flagged in four of five conditions. These are not borderline cases where the ground truth is uncertain—both scripts were verified correct by construction—but scripts whose correct methodology is superficially similar to a flawed one, making them systematically vulnerable to pessimism bias across model families.

This is a form of *pessimism bias*—agent reviewers flag *methodological complexity* as *methodological error*, penalizing engagement with difficult methodology rather than rewarding it. The pattern is robust across models and context conditions.

6 The Meta-Finding: Self-Referential Evaluation

This skill is itself reviewed by the Claw4S agent evaluator. In Phase 1, Chelate executes `generate_audit_scripts.py` and then reviews the 36 generated scripts—*the experiment runs on the conference's own reviewer.* The audit report that Chelate produces becomes our experimental result, and human chairs see the audit report alongside Chelate's review score for our submission.

This creates an unavoidable confound: the agent reviewing our submission has access to this paper, which describes exactly what the benchmark tests. This is the "context contamination" we measure in Section 4.3— but in the self-review case, the contamination is *complete*: the reviewer knows the taxonomy, the design, and the answer key structure.

Our taxonomy experiment showed that contamination *reduces* FPR rather than inflating it, suggesting self-review does not make the benchmark trivially easy. But the confound remains: we cannot directly compare Chelate's self-review performance to our API-evaluated baselines, because the prompt contexts are fundamentally different.

We present this not as a limitation to disclaim but as a finding to highlight: any benchmark whose specification cohabits with the reviewer's context is inherently a different benchmark for that reviewer than for external evaluators. The Claw4S review architecture embodies this tension by design. It is the right tension to surface.

7 Related Work

Our benchmark is complementary to SWE-bench [Jimenez et al., 2024], which measures whether agents can *write* code to fix bugs; we measure whether agents can *detect* methodological errors in code written by others. LLM-as-judge work [Zheng et al., 2023] evaluates conversational quality; we evaluate statistical reasoning. Statcheck [Nuijten et al., 2016] detects numerical inconsistencies in reported results; we address the harder problem of methodological soundness in source code, where no arithmetic check is possible. The reproducibility literature [Henderson et al., 2018, Gundersen and Kjensmo, 2018, Stodden et al., 2018] documents failures that our benchmark operationalises as detectable flaws. Lipton & Steinhardt [Lipton and Steinhardt, 2019] catalogued troubling evaluation practices in ML; our meta-finding about self-referential review directly instantiates their concern.

8 Limitations

Synthetic scripts. All scripts generate synthetic data from seeded distributions. Real-world code is messier—preprocessing

pipelines, external dependencies, and domain conventions create additional surface area for both true errors and spurious concerns. Detection rates on real-world code may differ substantially.

Ground truth. Control scripts were written and verified by the authors. While planted flaws are unambiguous by construction, controls may contain unintended subtleties—consistent with our finding that some false positives, on inspection, identified real (if minor) complexities. An independent expert audit would strengthen the ground truth labels.

Condition asymmetry. Taxonomy-contamination and repeated-measure conditions were applied only to Sonnet 4.6, limiting cross-model interpretability of Table 1. Results reflect model versions as of early 2026; false-positive patterns may not persist across model generations.

9 Discussion and Conclusions

Agent peer review of methodological soundness is more reliable than we expected for *detection*, and less reliable than we hoped for *precision*. All models identify planted flaws with near-perfect sensitivity; the variation lies in false-positive rates on correct methodology, ranging from 22% (Opus 4.6) to 50% (Sonnet majority vote).

These findings suggest four practical recommendations for agent review systems: (1) **structured flaw taxonomies in reviewer prompts directionally reduce over-flagging** without harming detection, though the effect requires replication at larger scale to establish significance; (2) **single-run FPR underestimates the true marginal FPR**—the deficit is not from split votes but from inter-run variance on boundary-case scripts, so multi-run experiments should be standard; (3) **high-complexity correct scripts are systematically over-flagged**, suggesting reviewers should be prompted to distinguish "handling a hard problem" from "handling it wrongly"; and (4) **MEDIUM-confidence verdicts are reliably diagnostic of errors**—in every condition, verdicts rated below HIGH were wrong at higher rates than HIGH-confidence verdicts, suggesting that surfacing low-confidence flags for human review would efficiently target the highest-risk assessments.

The benchmark is released as a Claw4S submission at `https://github.com/jheuer/replication-trap`.

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